CHURCH OF CHRIST INTERNATIONAL, ACCRA

Accra Central Branch • Financial Administration Office

OFFICIAL USER & OPERATIONAL MANUAL

Offline-First Progressive Web App for Church Fund Accounting

Base Currency:	Ghana Cedis (GHS) with USD Reporting View
System Architecture:	Offline-First Local SQLite with PowerSync & Supabase
Target Audience:	Treasurers, Pastors, Elders, Financial Secretaries, Counting Volunteers
Software Release:	Release 1.0 • November 2026 Production Edition

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Section 1: Introduction, Accounting Model & Architecture

The CCIA Church Accounting System is an enterprise-grade, offline-first Progressive Web App (PWA) designed specifically for the financial administration of our local congregation in Accra, Ghana. It provides rigorous financial governance, segregation of duties, multi-fund tracking, and full compliance with standard double-entry bookkeeping.

Key Architectural Principles:

- **Base Currency in Ghana Cedis (GHS):** All primary ledger accounts, cash on hand, bank accounts, and payroll records are denominated in GHS pesewas. Foreign currency receipts are stored with original currency and exchange rate.
- **USD Denominational Reporting View:** The denomination requires an executive USD summary. Rather than running an error-prone second set of books, the system converts GHS statements at the prevailing period closing rate or average rate.
- **True Offline-First SQLite Engine:** Church operations frequently encounter unreliable internet in Accra. The system embeds a complete SQLite database directly on each client device. Queries execute in under 15ms offline.
- **Zero-Trust Server-Side Enforcement:** PostgreSQL database triggers and deferred constraints reject unbalanced postings, unauthorized fund overdrafts, or postings into closed periods.

[ACCOUNTING PRINCIPLE] All monetary values are calculated in integer minor units (pesewas/cents) using Banker's Rounding (Round to Nearest Even) to eliminate 1-pesewa calculation drift.

Section 2: User Roles, Permissions & Security Hardening

The system enforces strict Segregation of Duties. No single individual can count cash, authorize expenses, and disburse church funds alone.

User Role	Permitted Responsibilities	Giving Privacy
Pastor / Lead Minister	Approve expense requisitions, inspect budgets, review high-level church financial statements.	Names Masked (••••••••)
Treasurer	Disburse payment vouchers, calculate Ghana WHT, manage bank accounts, enter FX rates, lock accounting periods, run payroll.	Full Access
Finance Elder / Auditor	Complete read-only audit access across all journal entries, counting sheets, bank reconciliations, and TB Doctor.	Full Access
Financial Secretary	Manage 300 member profiles, envelope numbers (ENV-001 to ENV-300), pledge fulfillment, and tax statements.	Full Access
Counting Team (Volunteers)	Record Sunday service physical cash piece counts, cheque tallies, MoMo float reconciliations, dual sign-off.	Names Masked (MEM-***)
Department Head	Submit expense requisitions for youth, children, evangelism, media, or welfare against approved department budgets.	No Giving Access

Security PIN & Lockout: The screen automatically locks after 5 minutes of inactivity. Entering an incorrect PIN 5 consecutive times triggers a 30-second lockout cooldown, defeating automated brute-force attacks.

Section 3: Executive Financial Administration Dashboard

The Dashboard presents four critical areas:

- **Executive KPI Cards:** Total Giving & Income (with converted USD badge), Total Expenditures (including Ghana WHT), Net Operating Surplus (with margin status), and Liquid Cash & Bank reserves.
- **Fund Integrity & Restricted Capital Protection:** Dynamic progress bar showing Unrestricted (General Fund) vs Restricted (Building, Missions, Welfare) capital, preventing illegal diversion of donor-restricted gifts.
- **Individual Fund Breakdown:** Live balances for General [GEN], Building [BLD], Missions [MIS], Welfare [WEL], and Youth [YTH].
- **Quick Operations Launchpad:** Fast shortcuts for Counting Sheets, Payment Vouchers, Financial Reports, and Ledger Integrity Diagnostics.

Section 4: Sunday Giving & Dual-Custody Counting Sheets

Sunday collections follow a strict dual-custody procedure:

- **Step 1: Enter Secure Room:** Following the benediction, two unrelated counting team members enter the counting office and lock the door.
- **Step 2: Physical Piece Count:** Counters tally physical cash by individual denomination. For Ghana Cedis: GHS 200, 100, 50, 20, 10, 5, 2, 1, and 50 pesewas coins. For US Dollars: USD 100, 50, 20, 10, 5, 1. The software calculates exact totals automatically.
- **Step 3: Cheques & Mobile Money:** Enter all cheques with Drawer Name, Bank, Cheque Number, and Amount. Record MoMo collections (MTN MoMo, Telecel Cash, AT Money) with Transaction Reference IDs.
- **Step 4: Fund Allocations:** Specify how the total collection divides across church funds: General Fund (tithes and regular offering), Building Fund, Missions, and Welfare. Total allocations must exactly equal the cash + cheque + MoMo total.
- **Step 5: Dual Sign-Off:** Counter #1 signs on-screen and enters their PIN. Counter #2 independently verifies the cash, co-signs, and enters their PIN. The system blocks the same user from signing both slots.

[AUTOMATIC POSTING] Co-signing the sheet moves it to POSTED, instantly creating balanced journal entries and generating an official Bank Deposit Slip.

Section 5: Expenditure, Requisitions & Payment Vouchers (GRA WHT)

Ghana Revenue Authority Withholding Tax (WHT) rates are computed automatically:

Category	GRA WHT Rate	Accounting Treatment
Professional Services & Contractors	7.5% WHT	Debit Expense for Gross, Credit WHT Payable (2020) for 7.5%, Credit Bank for 92.5%
Goods, Supplies & Equipment	3.0% WHT	Debit Expense for Gross, Credit WHT Payable (2020) for 3.0%, Credit Bank for 97.0%

Petty Cash / Internal Reimbursements	0.0% (Exempt)	Debit Expense for Gross, Credit Cash/Bank for 100%

Section 6: Church Staff Payroll & Ghana Statutory Taxes

Monthly payroll complies fully with Ghana labour regulations:

- **SSNIT Tier 1 (13.5%):** Employer mandatory statutory contribution remitted to SSNIT.
- **SSNIT Tier 2 (5.5%):** Employee mandatory contribution deducted from gross stipend and remitted to the occupational scheme trustee.
- **GRA PAYE:** Calculated using GRA's progressive tax bands. Produces printable PDF payslips for pastors and office staff.

Section 7: Member Giving Directory, Pledges & Official Statements

Confidential member directory tracking 300 church members with envelope numbers (ENV-001 to ENV-300).

- **Pledge Tracking:** Monitors dedicated capital campaigns (e.g. Building Fund Expansion) with real-time progress bars.
- **Tax Statements & Receipts:** Generate official annual PDF statements bearing church letterhead and signature blocks offline.

Section 8: Month-End Closing & Foreign Exchange (USD) Revaluation

Month-End Routine Steps:

1. Run Trial Balance Doctor to confirm books are green.
2. Enter official Bank of Ghana month-end closing exchange rate.
3. Run Revaluation Preview to calculate unrealized FX gain or loss on USD holdings.
4. Post revaluation entry to Account 5980 Foreign Exchange Gain/Loss.
5. Lock the accounting period to permanently freeze the month against changes.

[IMMUTABILITY RULE] Closed periods cannot receive new postings. Corrections must be posted in the open active period via a linked reversal.

Section 9: Trial Balance Doctor & Ledger Integrity Diagnostics

The Doctor runs 10 sequential diagnostic tests:

- **Rule 1: Mathematical Balance:** Verifies that total church-wide Debits equal total Credits in GHS.
- **Rule 2: Fund Balance:** Ensures that every fund (General, Building, Welfare, Missions) balances to zero independently.
- **Rule 3: Structural Integrity:** Scans for orphaned lines, headers without lines, or zero/negative amounts.
- **Rule 4: Sync Integrity:** Detects partial sync batches where lines present differ from header line_count.
- **Rule 5: Currency & Rounding:** Verifies base GHS equals USD x rate, flagging 1-pesewa rounding variances.

- **Rule 6: Duplicate Postings:** Flags duplicate transactions with matching date, amount, fund, and user within 15 minutes.
- **Rule 7: Pattern Detection:** Tests the remaining difference D: checks transposed digits (D divisible by 9), inverted debit/credit (D / 2), and decimal slips (10x, 100x).
- **Rule 8: Time Bisect:** Pinpoints the exact calendar date when the ledger first went out of balance.
- **Rule 9: Source Breakdown:** Groups differences by user, physical device prefix, and document type.
- **Rule 10: Opening Balances:** Verifies that previous fiscal year carry-forward balances balance independently.

[NEVER AUTO-FIX] Strict Never Auto-Fix Policy: Silent corrections undermine audits. The Doctor presents evidence and recommends linked reversal entries.

Section 10: Offline Operation, Network Syncing & Encrypted Backups

- **Offline Autonomy:** All transactions queue locally on SQLite. When reconnecting to Wi-Fi/MoMo, PowerSync uploads queued changes automatically.
- **Encrypted Backups (.ccbk):** Treasurers can download an AES-256 encrypted backup archive with a passphrase. In disaster scenarios, this bundle restores all ledgers and members completely.

Section 11: Demo Mode (300 Members & 26 Weeks Historical Data)

- **300 Ghanaian Members:** Authentic Akan, Ga, Ewe, Northern names, envelope numbers ENV-001 through ENV-300 across 8 ministries.
- **26 Weeks of Worship Services:** Balanced Sunday collections, cheques, MoMo, building pledges, payroll runs, and utilities.
- **Zero Risk to Live Books:** All demo operations run in an in-memory sandbox without touching live church records.

Section 12: Practical Step-by-Step Scenarios & Frequently Asked Questions

Q: What should I do if the Trial Balance Doctor displays RED?

A: Click 'Check & Diagnose Books' to read the Plain-Language Report. Check Rule 7 to see if a digit transposition occurred (e.g. entering 450 instead of 540) or if a debit was posted as a credit. Use the link to open the flagged entry and click 'Create Linked Reversal Entry' to mirror and correct it.

Q: Why does the system block me from signing both counting sheet slots?

A: Internal control policies require two distinct, unrelated individuals to verify and sign Sunday cash counts. The second counter must log in with their own credentials.

Q: How do we handle foreign currency donations given in US Dollars?

A: Record the USD amount under the USD cash denomination section. The system records the original USD amount and converts it to GHS using the active exchange rate. At month-end, run Foreign Currency Revaluation to recognize any exchange gains or losses.

Q: What if our church laptop's screen is locked and the PIN is entered wrong 5 times?

A: The keypad locks for 30 seconds to prevent brute-force attacks. Wait 30 seconds for the cooldown timer to expire, then re-enter the correct PIN.